

Newts and salamanders

Order Caudata

Greater siren

Siren lacertina

- ↔ 20–35 in (50–90 cm)
- 📍 E. and SE. US
- ⊗ Locally common

This large salamander of the lower Mississippi valley lacks hindlimbs and the entire pelvic girdle. It has tiny front feet and can move through the water with eel-like undulations of the body. Larval gills are retained into adulthood, growing large and feathery as the animal matures. The siren rarely

feathery,
external
gills

leaves the water, except during heavy rains—but can survive drought by cocooning in mud.



Hellbender

Cryptobranchus alleganiensis

- ↔ 12–29 in (30–74 cm)
- 📍 E. US
- ⊗ Near threatened

America's largest salamander is a member of the same family as the Oriental giant salamanders, the biggest in the world. It rarely leaves the water. Adults have lungs, but in their oxygen-rich, fast-flowing river habitat, gaseous exchange is supplemented by absorption of oxygen across the flabby skin. The male uses his flat head to excavate a protective cavern beneath rocks, in which he fertilizes the female's eggs and guards them until they hatch.

Mudpuppy

Necturus maculosus

- ↔ 12–19 in (30–48 cm)
- 📍 S. Canada, C. and E. US
- ⊗ Common

The mudpuppy is a water-breathing salamander—its feathery gills become bigger in stagnant habitats with lower oxygen concentration. An aggressive, nocturnal predator of small animals, it may become active by day when the water is muddy. Unlike many other exclusively aquatic salamanders, which disperse sperm into the water, the mudpuppy reproduces like newts—males produce a spermatophore (sperm package) for transferring sperm to the female's body.

Fire salamander

Salamandra salamandra

- ↔ 7–11 in (18–28 cm)
- 📍 Europe
- ⊗ Common

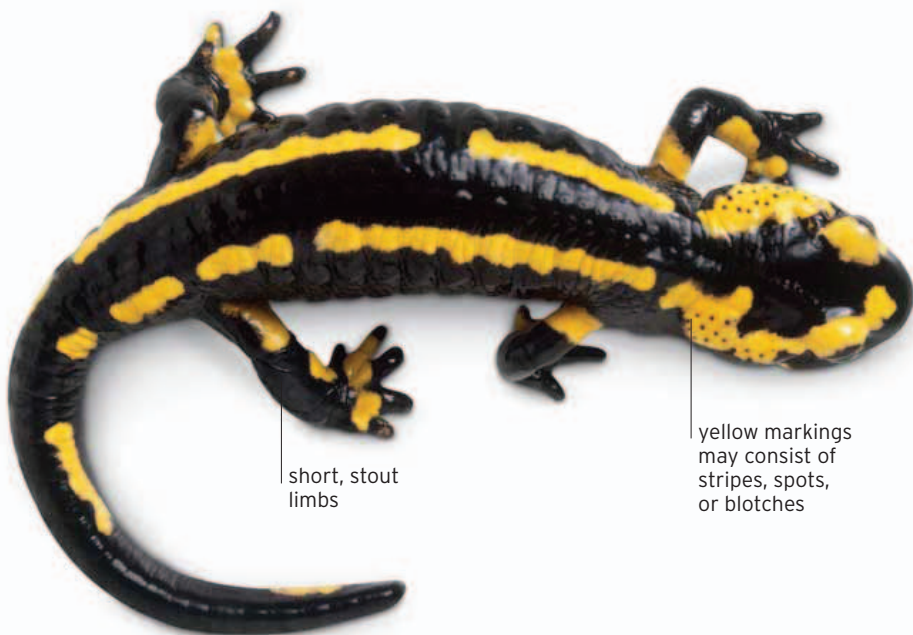
The bright warning spots of this woodland salamander mark the locations of glandular patches of skin that secrete toxic secretions, which are used in defense against predators. Fire salamanders typically develop from aquatic larvae. Adults are almost entirely terrestrial, except when breeding; however, some populations give birth to live young on land that are miniature versions of adults.

Alpine newt

Ichthyosaura alpestris

- ↔ 2–5 in (5–12.7 cm)
- 📍 Europe
- ⊗ Not known

A typical newt, this amphibian leads a dual aquatic–terrestrial life. Outside the breeding season, the adults live on land, moving into ponds to breed. Here, the males grow a dorsal crest and develop brighter colors to court females. Eggs hatch into larvae, which metamorphose into terrestrial adult forms called eft. Only southern populations are truly alpine in habitat.



short, stout
limbs

yellow markings
may consist of
stripes, spots,
or blotches

Eastern newt

Notophthalmus viridescens

- ↔ 3–5 in (7.6–12.7 cm)
- 📍 E. Canada, E. US
- ⊗ Common

After metamorphosing from the aquatic larval stage, this newt develops into a tiny, terrestrial, lizardlike eft with red spots. The pattern may act as a warning to predators that the skin of this amphibian is extremely toxic. The efts spend up to four years of their life on land in woodland, but return to pools to grow into larger breeding adults; they retain the red-spotted colors.

Mexican axolotl

Ambystoma mexicanum

- ↔ 4–8 in (10–20 cm)
- 📍 Mexico (Lake Xochimilco)
- ⊗ Critically endangered

Although the Mexican axolotl belongs to a family of mole salamanders that are typically terrestrial as adults, it remains aquatic and retains some larval characteristics into adulthood, including feathery gills and a finlike tail. Axolotls are popular pets, but the wild population—almost entirely confined to a lake near Mexico City—is critically endangered.

Eastern tiger salamander

Ambystoma tigrinum

- ↔ 7–14 in (18–35 cm)
- 📍 North America
- ⊗ Common

Named for its vivid skin pattern, this typical burrowing mole salamander usually spends its adult life on land. Adults breed in ponds—often returning to their place of birth to do so. Most larvae metamorphose into terrestrial adults, but under some conditions—such as in the cool Rocky Mountains—they may stay aquatic and retain gills right into sexual maturity.



short, powerful
limbs used for
burrowing

Other species

- **Ensatina**
Ensatina eschscholtzii » p59
- **Great crested newt**
Triturus cristatus » p170
- **Japanese giant salamander**
Andrias japonicus » p291